WP4 - Hysteresis Compensation

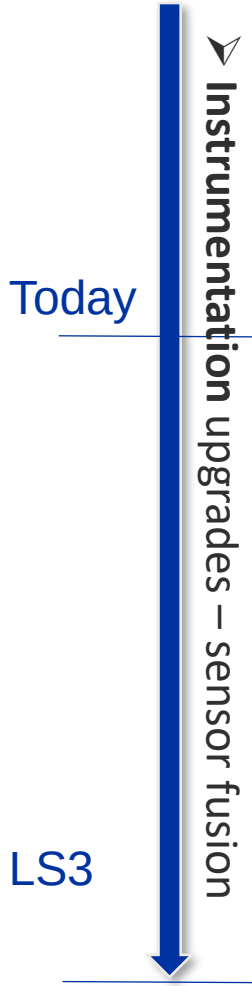
C. **Petrone,** B. Devlin-Hill, A. Bellelli, A. Ganesh, M. Bonora, A. Huschauer, F. Velotti, M. Schenk, A. Lu, V. Kain, K. Li

Thanks to:

R. Chritin, U. Hernandez of TE-MSC, R. Murillo Garcia of SY-EPC, N. Chritin, R. Betemps, A. Marichal and R. Valle of EN-MME

Indico link: 1719908

Status, ongoing and plans



➢ Achievements

- ✓ AI/ML lessons learned
- ✓ MD1 -> flattened
- ✓ Proof of concept: UCAP field compensator

➢ **Ongoing**

- ➢ Instrumentation upgrades – sensor fusion
- ➢ Hysteresis modeling and field prediction

➢ Targets for LS3

- ➢ X-Suite sensitivity studies and compensator
- ➢ Enhance real-time field measurements
- ➢ Hysteresis modeling and field prediction

Status before 2026 – ML Approach

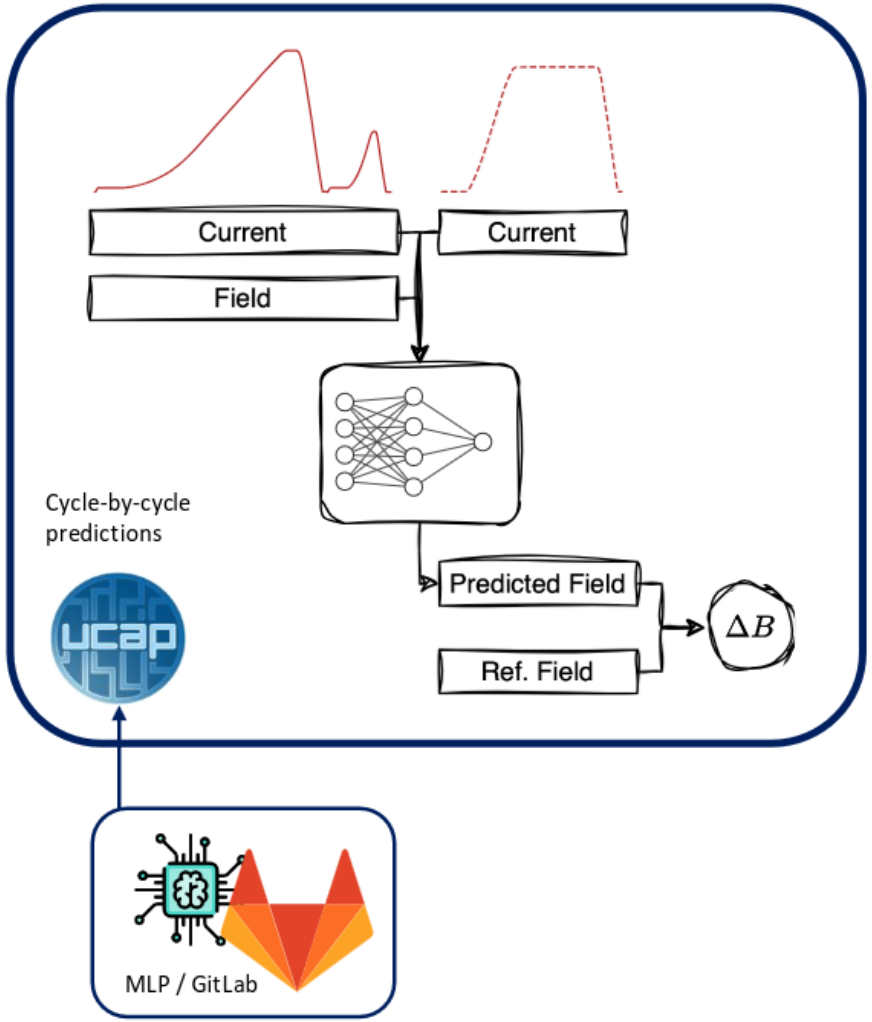
Neural Network to **predict** dipole hysteresis based on **complete** time series of history (from B-Train and Lab)

- Computationally very heavy
- Overfitted

ML Approach (Temporal Fusion Transformers) is not generalizable (**does not extrapolate**)

Black-box approach **ignoring the physics**

- Too sensitive to input data

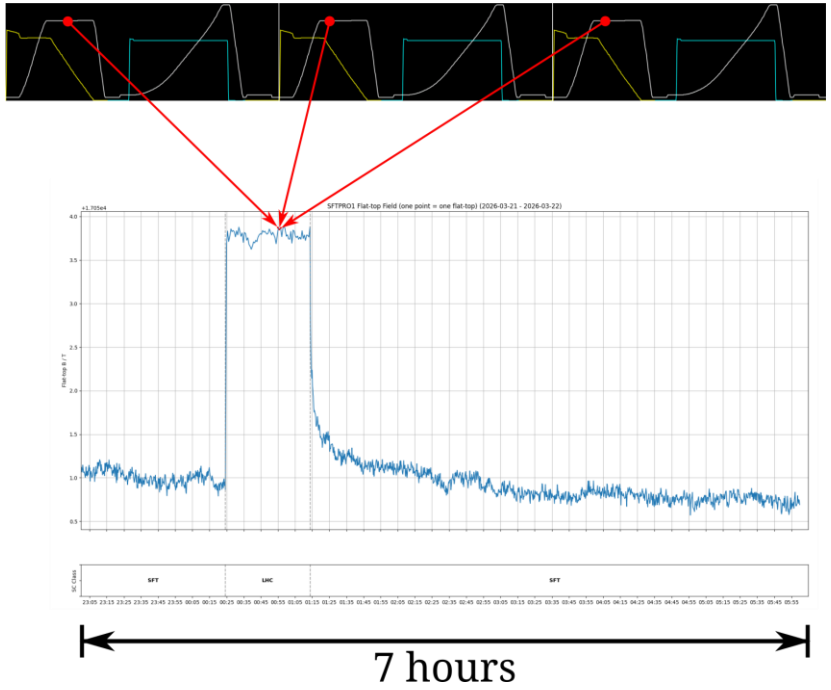


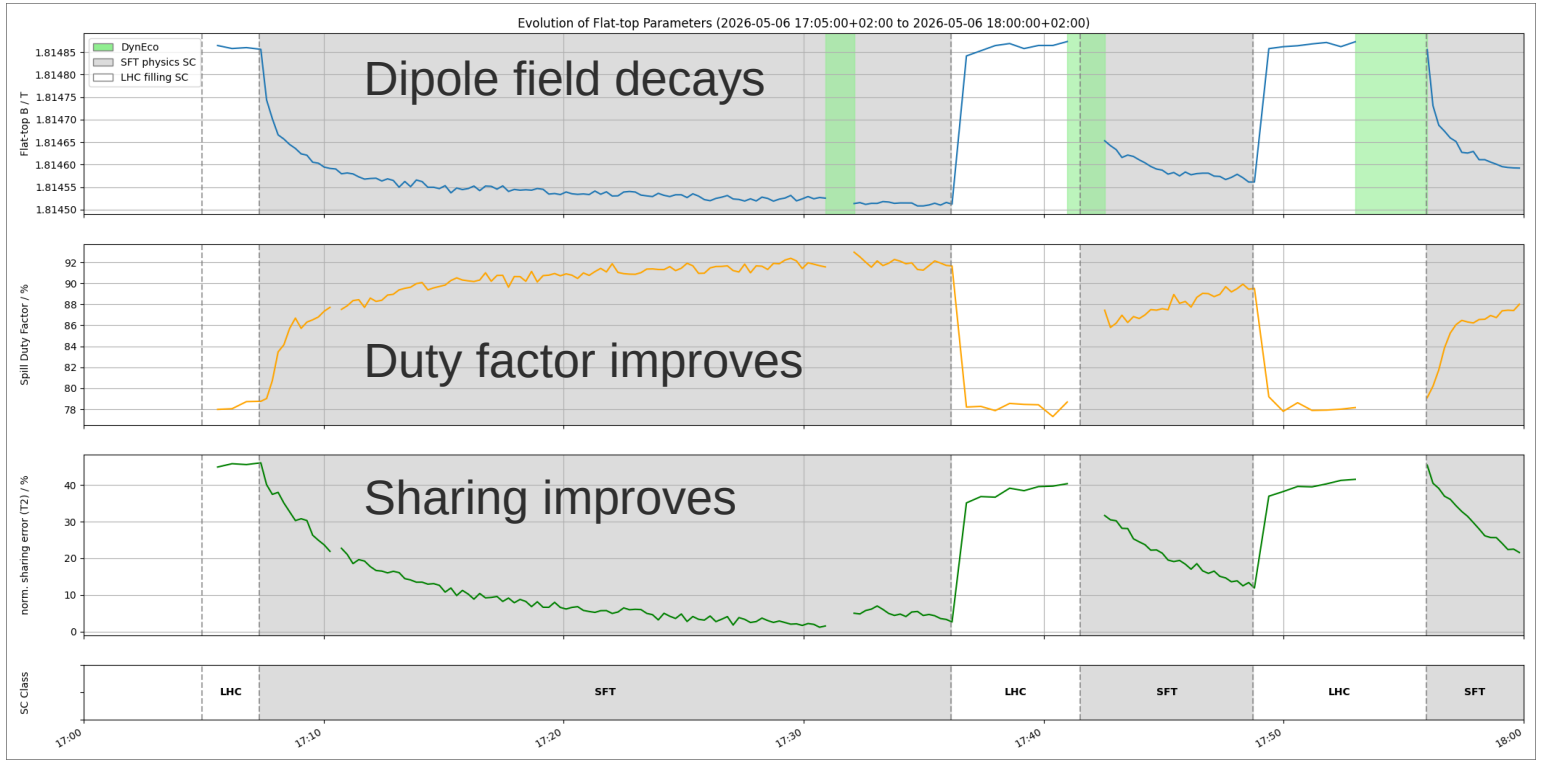
Slow SFTPRO1 Drifts

When switching SC from <u>SFT physics, to LHC filling, to SFT physics</u>, a slow drift of SFTPRO1 beam parameters is observed.

- **The flat-top dipole field drifts over hours**
- **<u>Can we fix the spill by eliminating this drift?</u>**

Grey: SFT physics SC
White: LHC filling SC





UCAP compensator

Very simple feed-forward compensator

- Reads **B-Train**, **compares to LSA-programmed B**, calculates difference, sends current trims

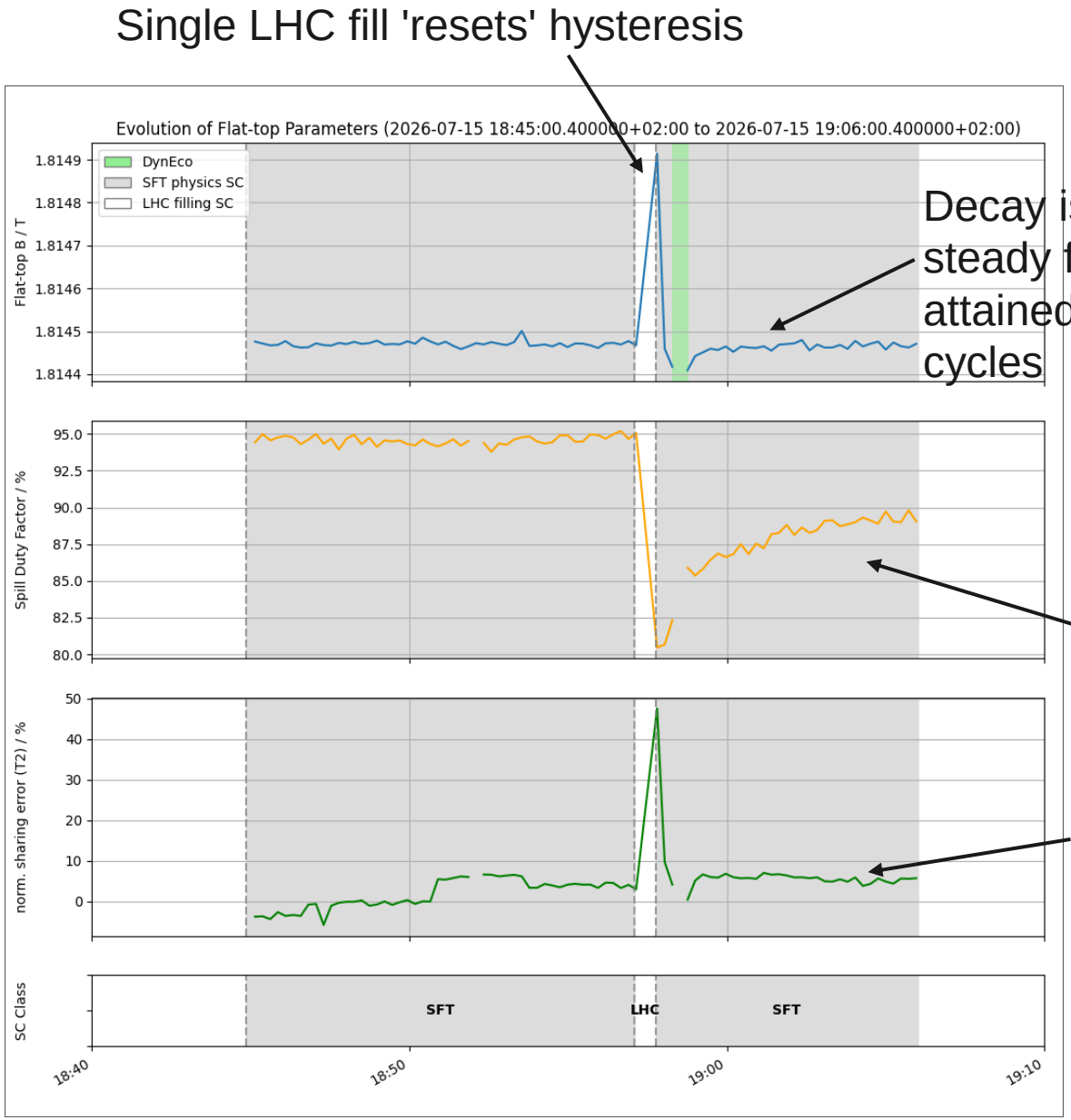
Works very well to keep field at flattop consistent

Struggles in vicinity of SC change

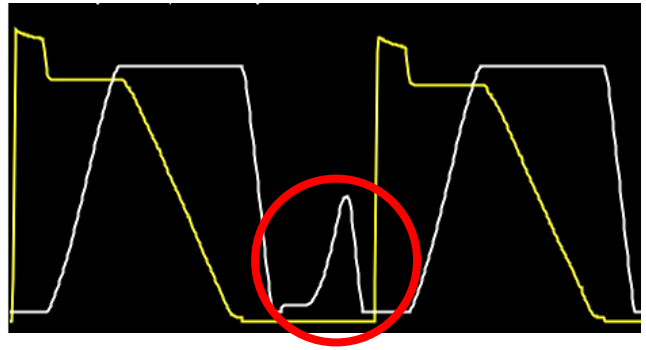
- Preliminary results using **light model** (fit) solves this – tests well on historical data

Operational tests:

compensator corrects sharing, but not spill



MD1 Flattened in 2026



MD1 ramp: **4.6 GWh/year, 300 kCHF/year**

MD1 (Expensive Cycle - No Beam)

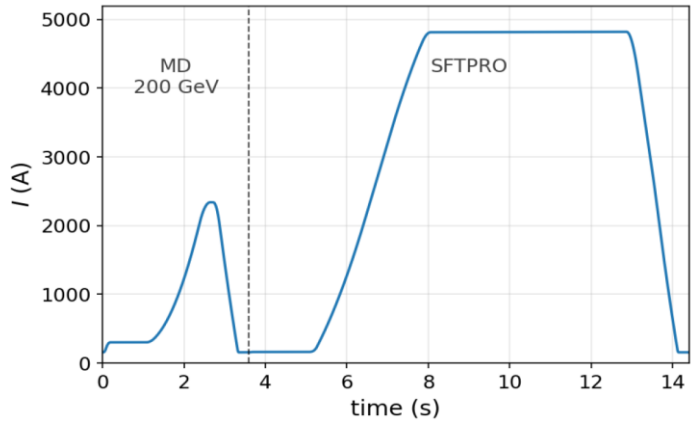
- Running Since ~**2006** (Beam Scheduling For CNGS)
- CNGS: **Stopped 2012**
- MD1 was thought to compensate for some hysteresis
- It was "Discovered": Not Needed For Hysteresis Comp.

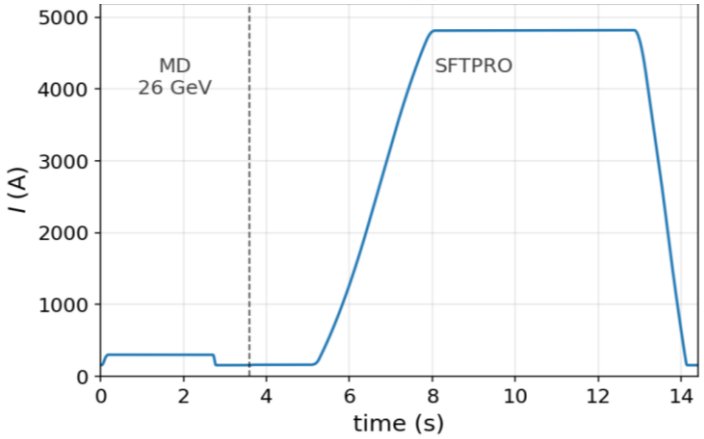
CHROMATICITY MEASUREMENTS

- Measured Chroma with Flat/Spiked MD1
- Measured chromaticity/tune match **LSA target** more closely
- Correction value in **LSA decreased**
- Chroma measurements confirmed MD1 could be removed

MD1 "flattened" in 2026 at 26 GeV

- In operation since **18th of February 2026**

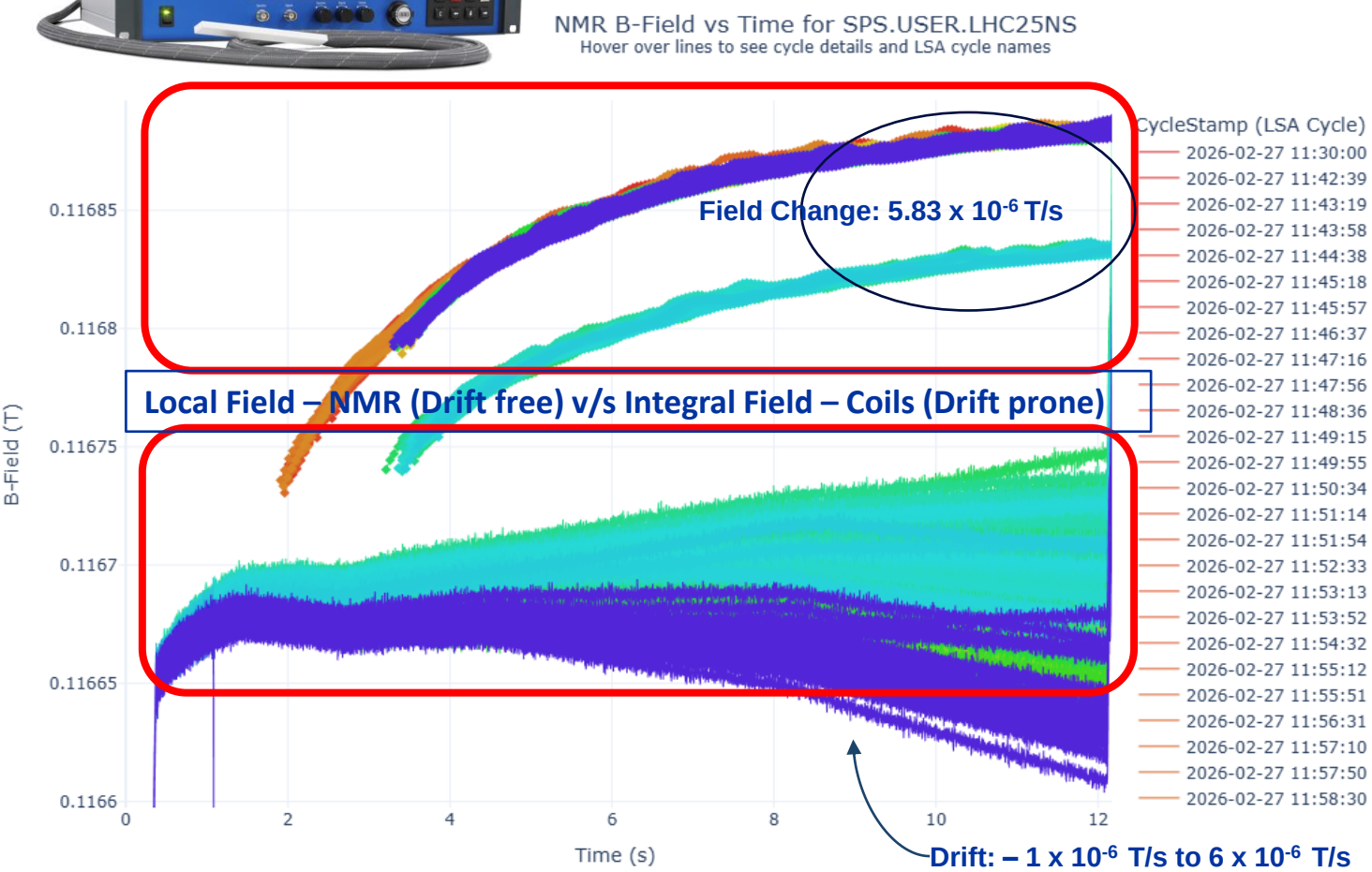


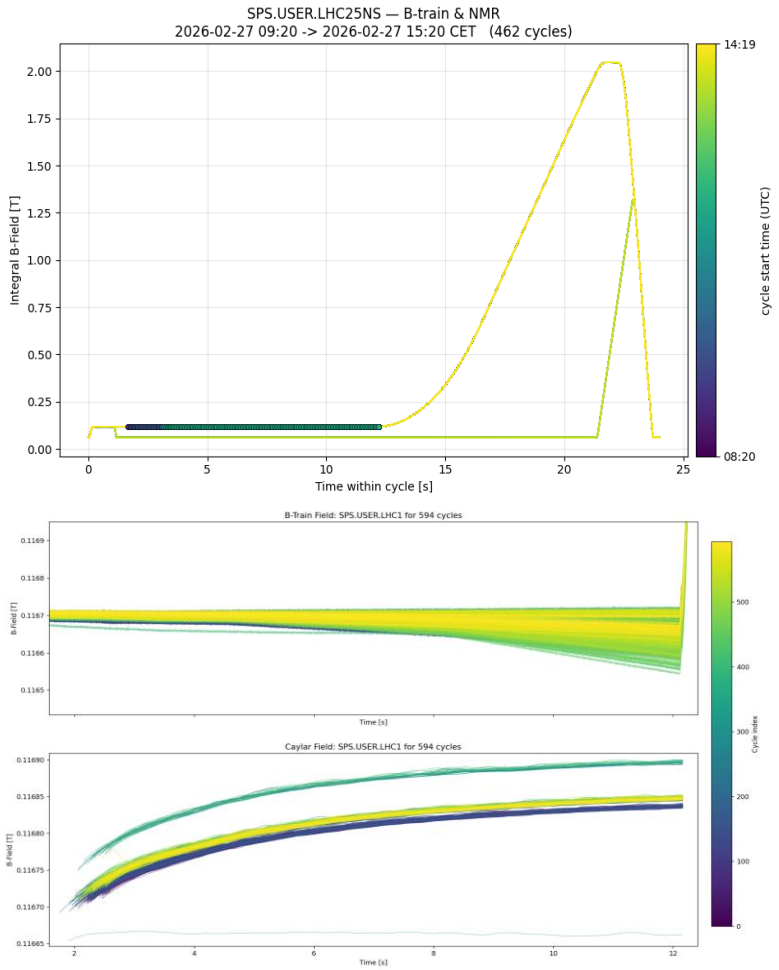


SPS B-Train instrumentation upgrades (2026)



Two NMR probes installed in the B-Train aperture (63 mT and 116 mT) - January 2026





Unknown position inside magnet → no direct integral field measurement

Study on the correlation of the NMR positions and field integral reconstruction during 2027

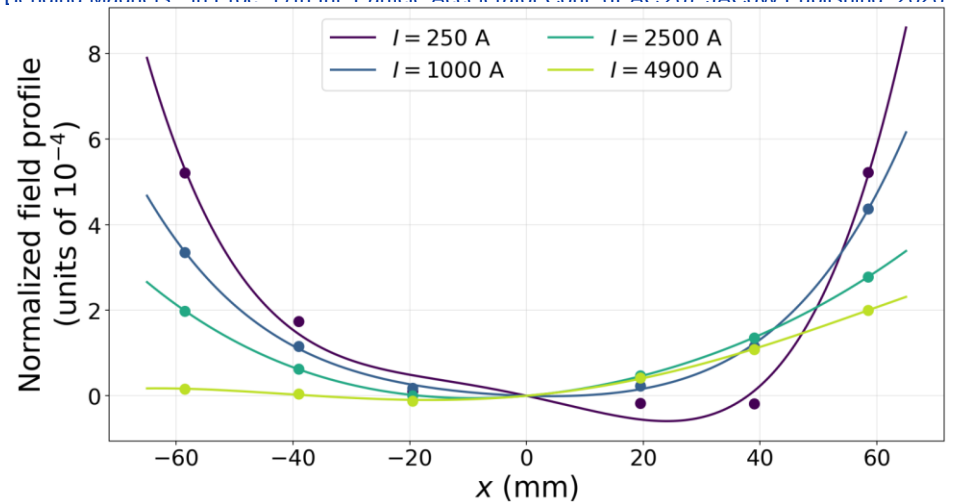
Laboratory instrumentation upgrades (2026)

A. Bellelli, A. Ganesh, C. Petrone, M. Buzio, M. Bonora-Tam, A. Huschauer, and S. Russenschuck, "Static and Dynamic Field Characterisation of the Super Proton Synchrotron Bending Magnets", in Proc. 17th Int. Particle Accelerator Conf. (IPAC'26), JACoW Publishing, 2026

- **New fluxmeters** (PCB-based coil magnetometers) for SPS magnets

- **MBA/MBB dipoles**
  - MBA fluxmeter prototype assembled, validated
  - Absolute field (equipped with NMRs)
  - Better transverse sextupolar resolution (40 % more)

- **QF/QD quadrupoles**
  - Prototype in operation
  - Absolute gradients (equipped with Hall sensors)









Targets for LS3

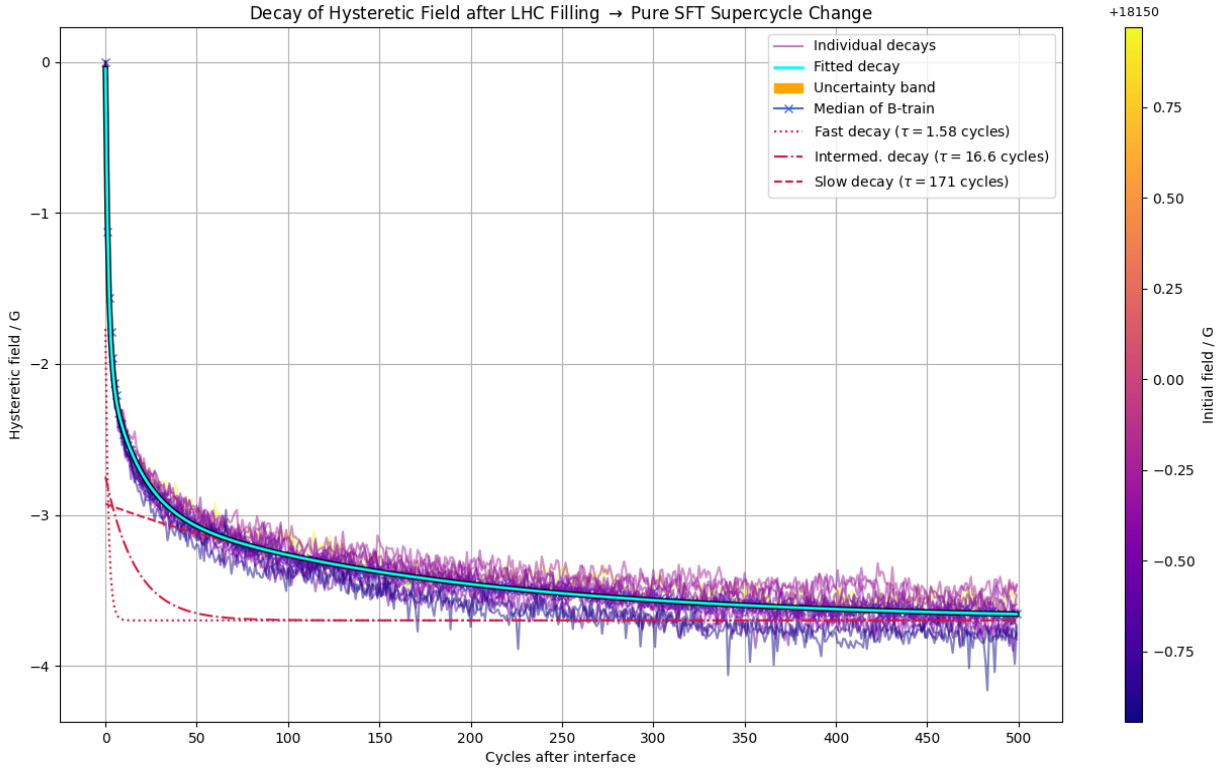
Targets for LS3 (Improved Compensator)

B-Train historical and laboratory data to develop improved dipole hysteresis compensator

- **Rules-based models** for SC changes
- Generalisable, yet **light predictive models** using only field extrema

Nice results already by fitting the slow SFTPRO1 decay to a triple exponential

- **Experiment** with more **generalised** approaches in LS3



Fit of a triple exponential of about a dozen LHC->SFT decays

Targets for LS3 (Beam Simulations)

Understanding hysteretic effects on the beam via xSuite

- Tune **drift studies** show that the **hysteresis effects** the beam in unexpected ways
- Use xSuite to probe relevant mechanisms
  - Radial/synchro-loops
  - Feed-down effects and impacts
  - Impact of the higher order multipoles of the dipoles (e.g.: sextupolar)

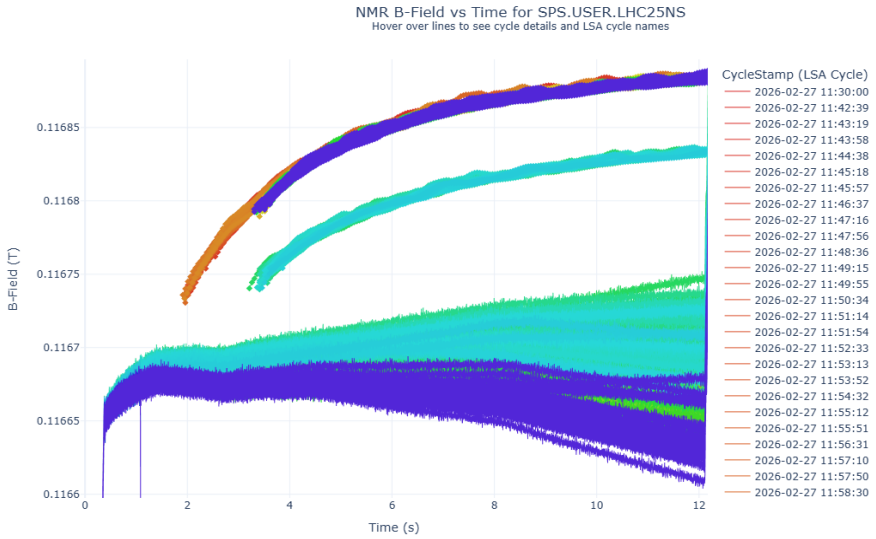
Integrate historical hysteresis data (NXCALs) to investigate effects on extraction, etc.

Use **xSuite** over LS3 to reproduce **observed behaviors** (quadrupole-tune coupling), conduct **sensitivity studies** (e.g., Gauss-level hysteresis in quadrupoles), and **integrate lab/historical data**.

Targets for LS3 (SPS B-train upgrade)

Upgrade of SPS B-Train fluxmeter for field control

- **Replace not operational NMR probes**
- **New Fluxmeters (integrated Hall probes and NMRs)**
- **Development in Electronics**
  - Hall probe control and readout
  - Sensor fusion
- **Development to send field to FGC**
  - White-rabbit receiver / adapter on FGC side
  - Study the reversibility for testing reason



- **Big investment**, hard to fall back (fluxmeter replacement, field control)
- **For full exploitation**, needs upgrade: sensor, electronics, FPGA, software, algorithms

Conclusion: upgrade and tests for LS3 (TE-MSC) in collaboration with NGC -> full upgrade for LS4

Targets for LS3 (B-train upgrade)

Upgrade of PS B-Train fluxmeter

- **LS3: Replace existing partial F and D fluxmeters with a full integral sensor**
  - **Possible to fall back to existing partial F and D fluxmeters**
  - **Works with existing B-Train HW**
- Integrate additional field sensors

- Run4: Spare coils + Hall sensors for development system
  - Testbench for NextGen B-train field measurement electronics
  - Extra acquisition and prototyping (easy magnet aperture access)

- LS4: Upgrade of all B-train **electronics**

TE-MSC /NGC

Targets for LS3 (B-train upgrade)

LS3: Monitoring, Fault Detection, Recovery

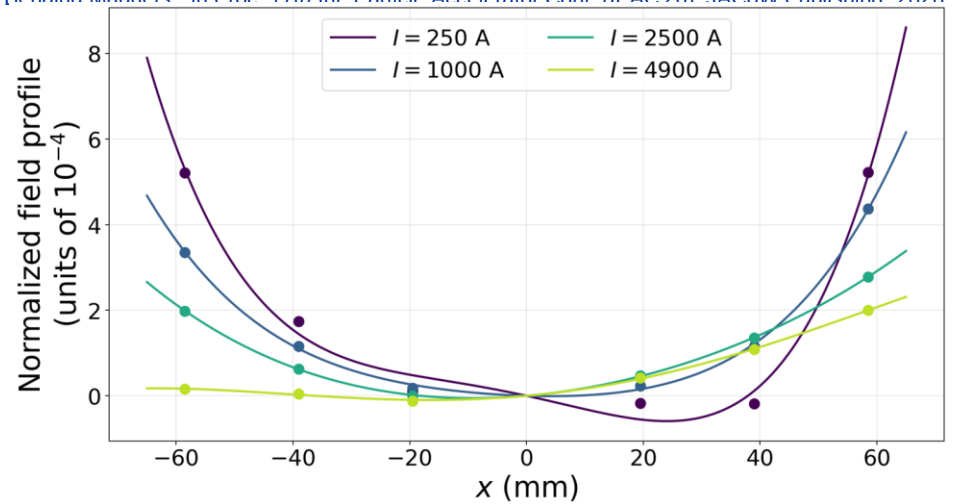
- Extend **Monitoring** and **Data collection** from running B-train systems
- **Fault detection** and **notification**, agent skill to integrate into analysis
- **Early degradation warning**, suggested tuning parameters (consistent OP/SP output)

--> Upgrade and tests for LS3 - TE-MSC with the help of WP8

Laboratory instrumentation for LS3

A. Bellelli, A. Ganesh, C. Petrone, M. Buzio, M. Bonora-Tam, A. Huschauer, and S. Russenschuck, "Static and Dynamic Field Characterisation of the Super Proton Synchrotron Bending Magnets", in Proc. 17th Int. Particle Accelerator Conf. (IPAC'26), JACoW Publishing, 2026

- **New fluxmeters** (PCB-based coil magnetometers) for SPS magnets
- **MBA/MBB dipoles**
  - MBA fluxmeter prototype assembled, validated
  - Absolute field (equipped with NMRs)
  - Better transverse sextupolar resolution (40 % more)
  - **MBB fluxmeter in advanced design phase**
  - **Production and assembly foreseen during 2027**
- **QF/QD quadrupoles**
  - Prototype in operation
  - Absolute gradients (equipped with Hall sensors)
  - **Optimization and manufacturing of three new units**
- **New measurement system** for multiple-channel DAQ acquisitions
- **Sensor fusion** methodologies and algorithms









Targets for LS3 (magnets modeling)

Build physics-based forward models of the SPS dipole

- **Physics-structured forward field model**, identified from measurements.
- The **parameters** are **identified** from a large set of measurements, including saturation curves, energy-based hysteresis models, eddy time constants, temperature, etc.
- **Status:**
  - A first model has already been developed for the NA magnet using induction coils and NMR measurements. Encouraging result 4×10^{-4} at flat tops.

Target: For LS3, proceed with modeling the SPS dipole at 10^{-5} at flat tops.

New EPA doctoral student - starts November 2026



home.cern

SPS Quadrupoles MEASUREMENTS

Quantification of the hysteresis of SPS quadrupole magnets

(prototype quadrupole fluxmeter)

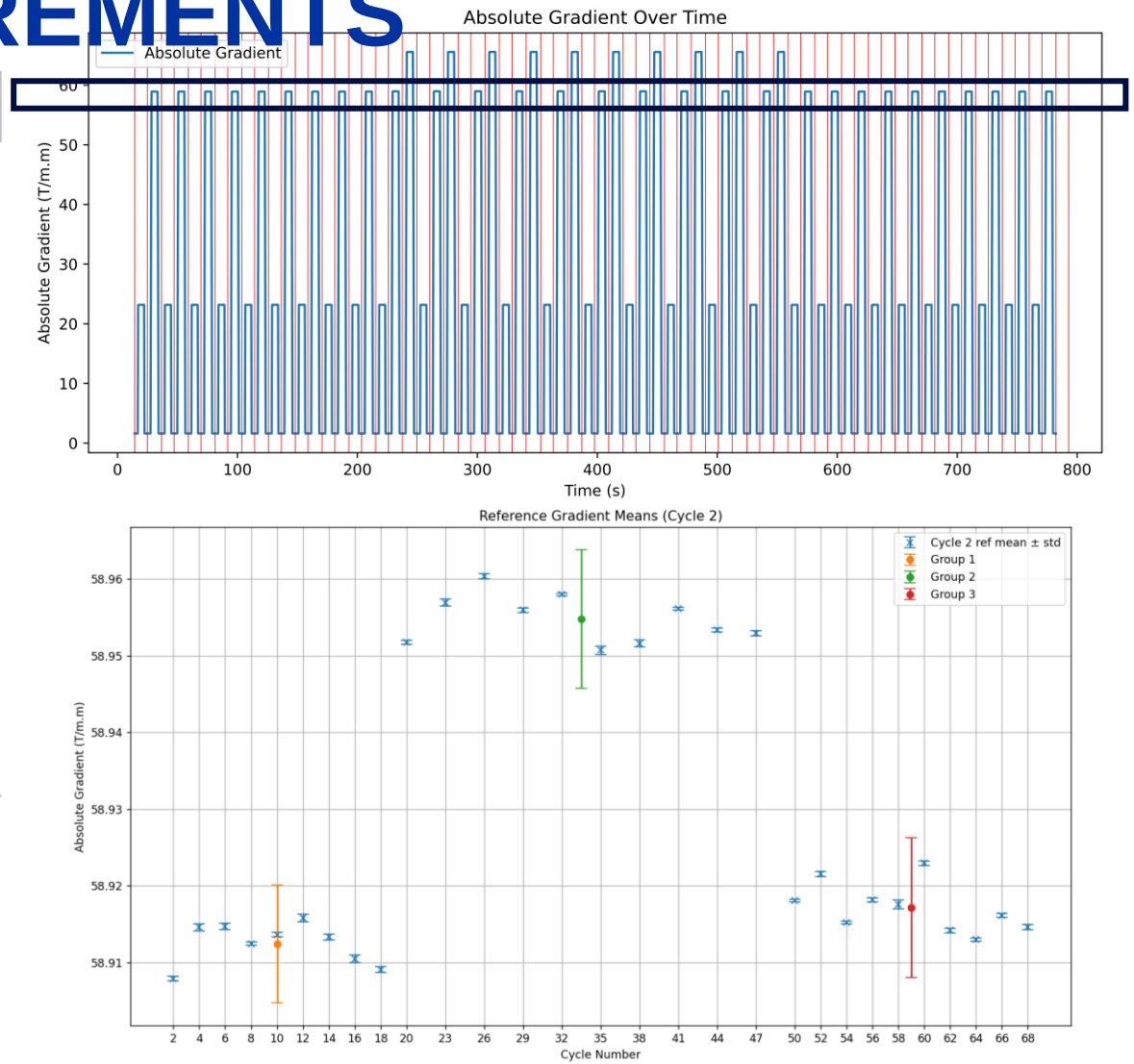
- Supercycle changes show a (measurable) difference in absolute field gradient
- Relative difference in absolute gradient that we can measure: 1.55 units (0.0091 Tm/m)

Challenges

- Fluxmeter alignment
- Integration drift correction
- Quadrupole marker accuracy

Upgrades for LS3

Development of a second iteration on-going. Design phase completed, manufacturing underway with EN-MME



Change of absolute gradient in the SFTPRO1 “flat-top” during ***MD-SFTPRO*** supercycle to the ***MD-SFTPRO-LHC*** supercycle

Absolute pulsed-mode, long-duration integral field instrumentation using multi-sensor approach (A. Ganesh, A. Bellelli, M. Bonora-Tam, M. Buzio, C. Petrone), doi: 10.18429/JACoW-IPAC2026-MOP7069

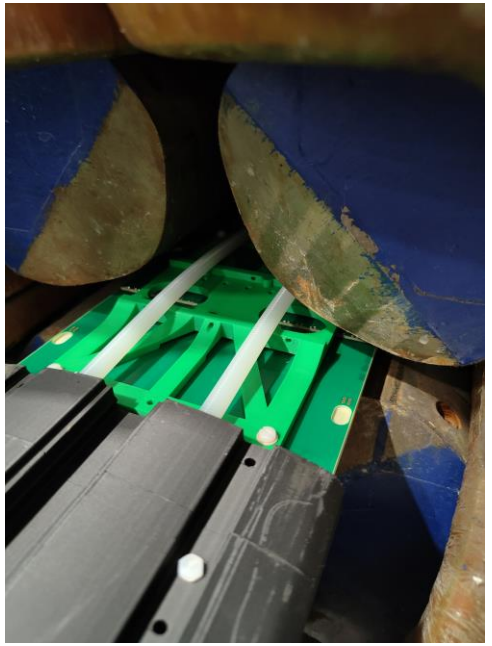
Instrumentation upgrades (2025)

- **New measurement racks** for the 867 laboratory
  - Implementation of online integration with DAQ systems
  - Replacement of the current SPS measurement system (FDI-equipped)
- **New fluxmeters** (PCB-based coil magnetometers) for SPS magnets
  - **MBA/MBB dipoles**
    - Design completed (EDA-04955-V1), delivery in September 2025
    - Next step: validation and calibration
    - Addition of NMRs for absolute local measurements
  - **QF/QD quadrupoles**
    - Design, development, and validation completed
    - First iteration of PCB fluxmeter is currently working
    - Addition of Hall sensors for absolute local measurements

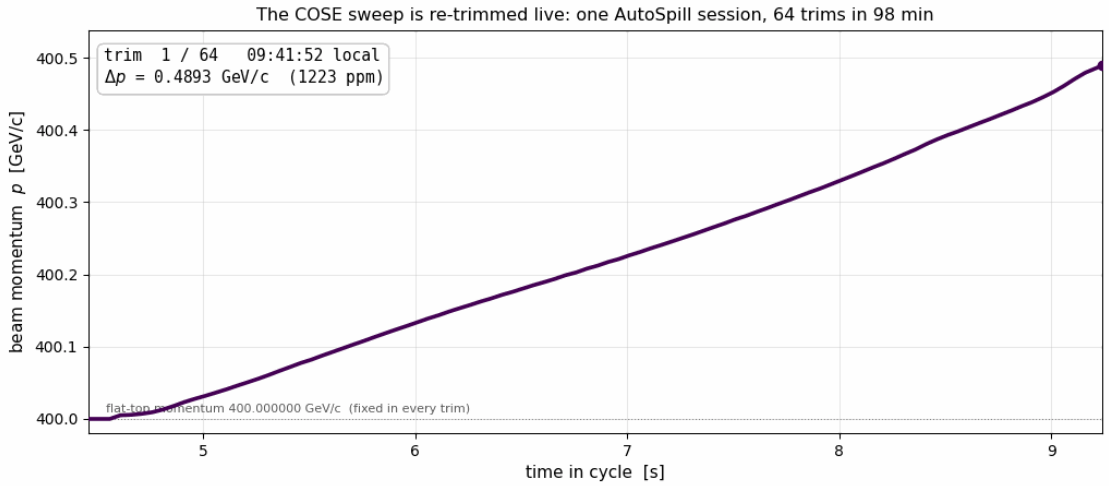


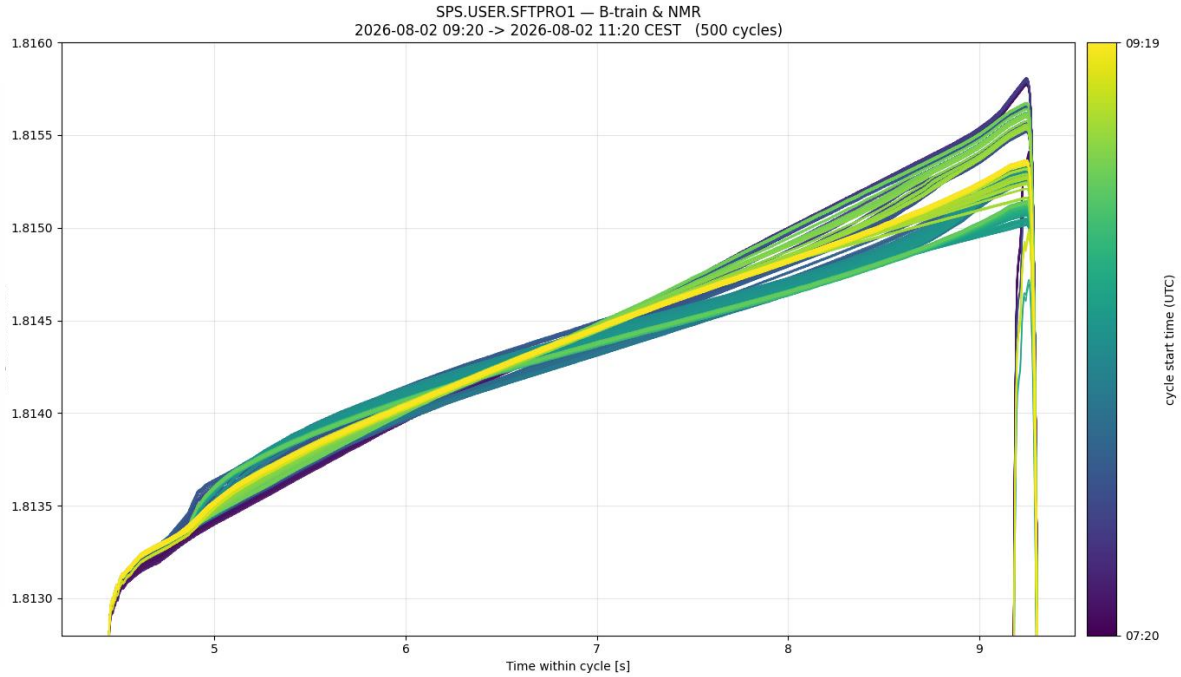


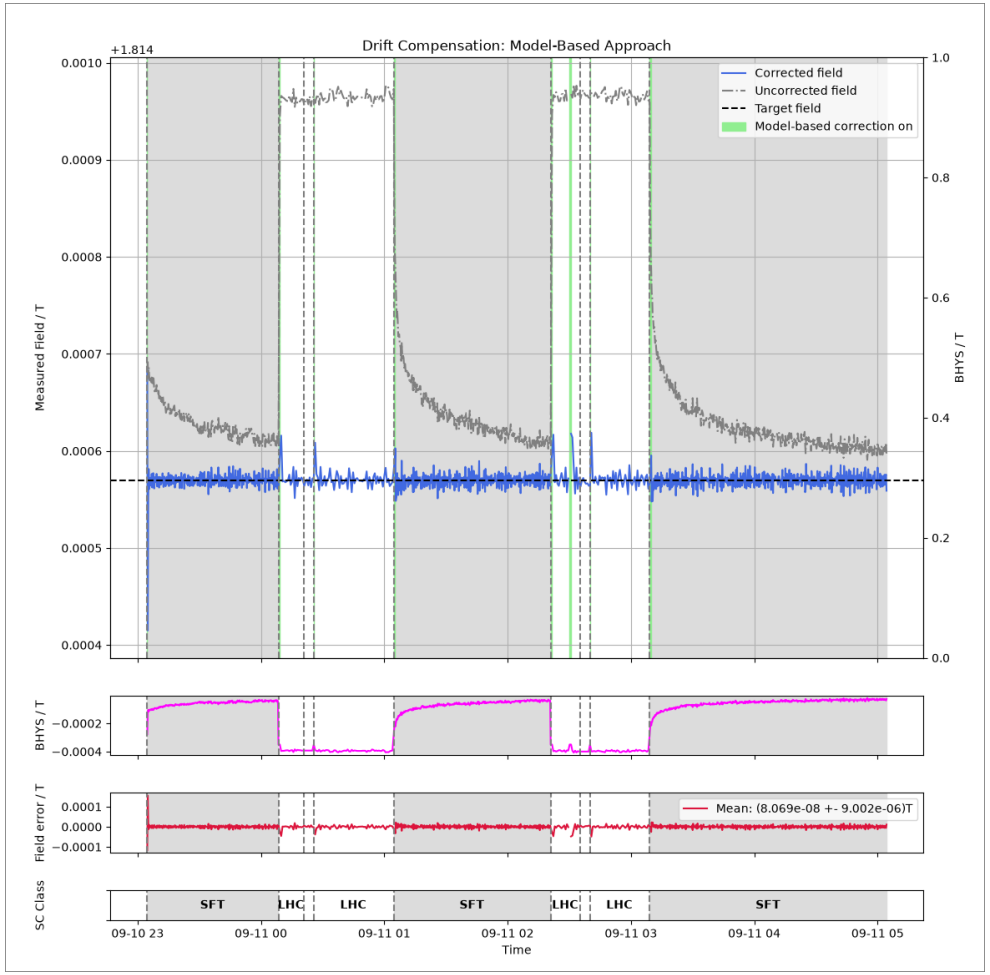


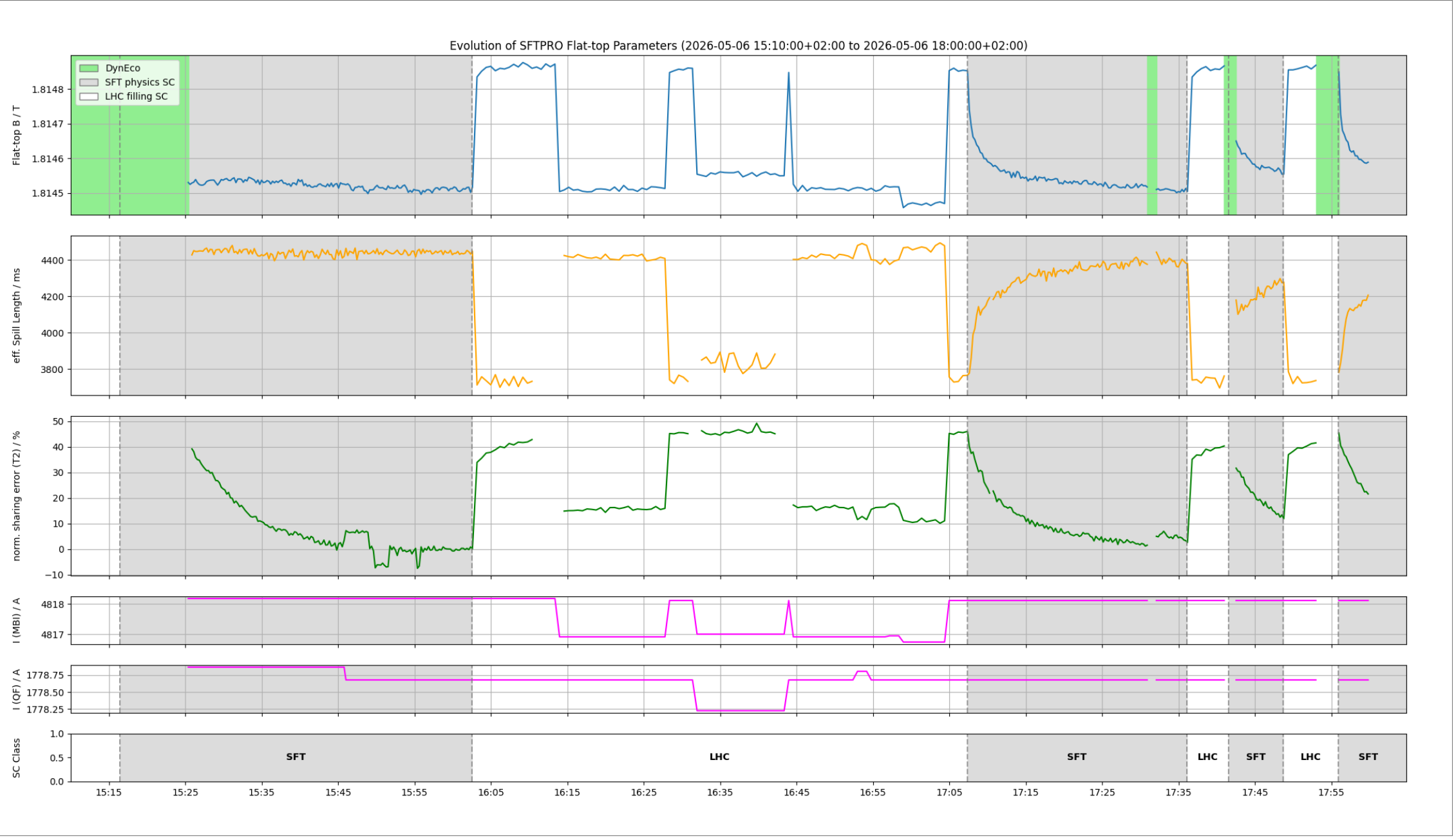


Bridging Beam-Operations and Magnetic Measurements





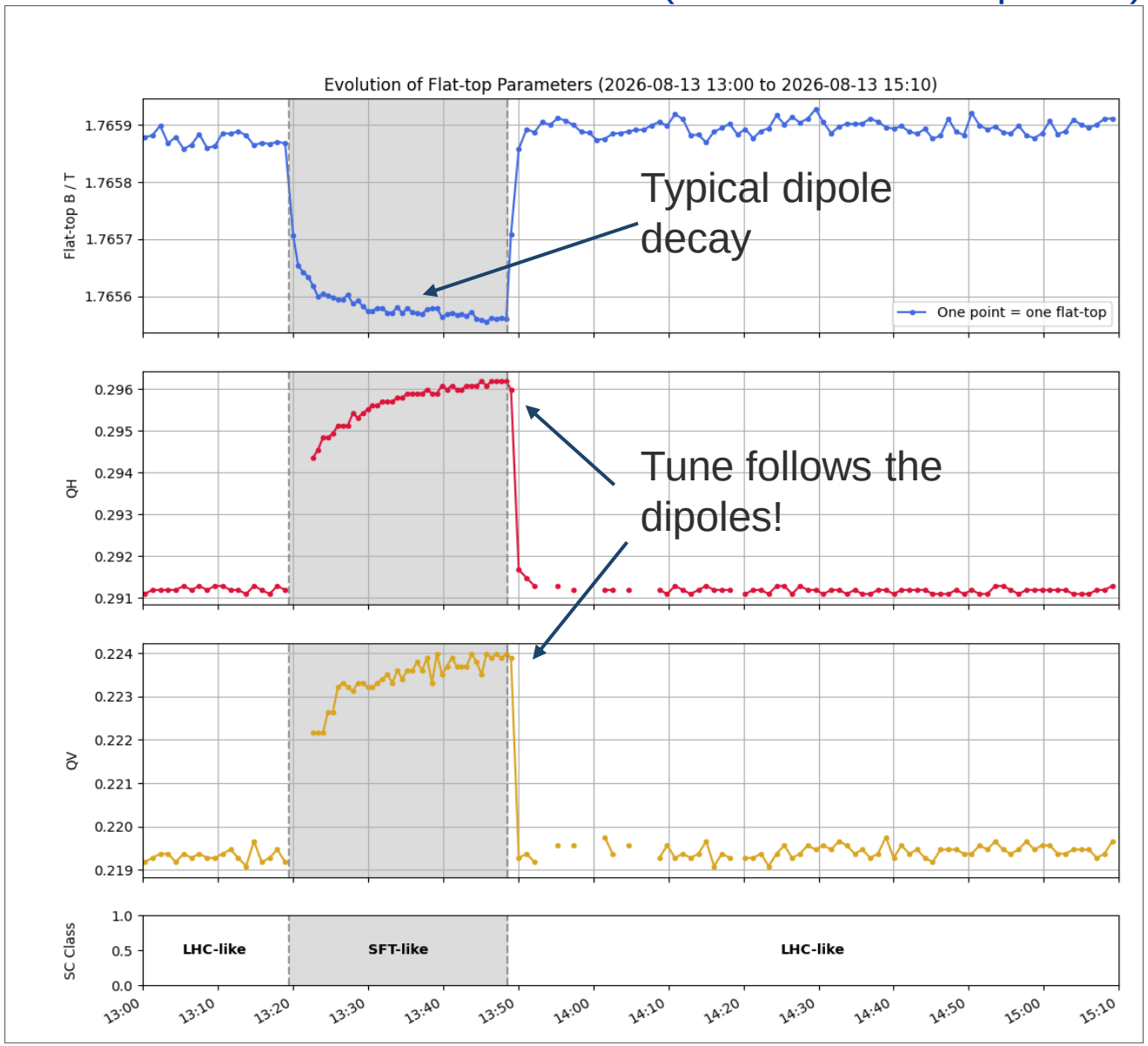

Drift Compensation: Model-Based Approach
+1.814
Corrected field
Uncorrected field
Target field
Model-based correction on
Measured Field / T
BHYS / T
0.0010
0.0009
0.0008
0.0007
0.0006
0.0005
0.0004
1.0
0.8
0.6
0.4
0.2
0.0
BHYS / T
−0.0002
−0.0004
Field error / T
0.0001
0.0000
−0.0001
Mean: (8.069e-08 +- 9.002e-06)T
SC Class
SFT
LHC
LHC
SFT
LHC
LHC
SFT
09-10 23
09-11 00
09-11 01
09-11 02
09-11 03
09-11 04
09-11 05
Time


Evolution of SFTPRO Flat-top Parameters (2026-05-06 15:10:00+02:00 to 2026-05-06 18:00:00+02:00)
DynEco
SFT physics SC
LHC filling SC
Flat-top B / T
eff. Spill Length / ms
norm. sharing error (T2) / %
I (MBI) / A
I (QF) / A
SC Class
SFT
LHC
SFT
LHC
SFT
LHC
SFT
15:15
15:25
15:35
15:45
15:55
16:05
16:15
16:25
16:35
16:45
16:55
17:05
17:15
17:25
17:35
17:45
17:55

Beam Measurements

(Below: Radial loop at FT)

- Compensating dipoles should let us examine pure quadrupole effects
- ...but tune appears to be coupled to dipole hysteresis
  - Unexpected: not (purely) a quadrupole effect
  - Radial loop suspected
- Ran with synchro-loop at FT and compensator on: appeared to restore both field and tune
- During ions: couldn't extract



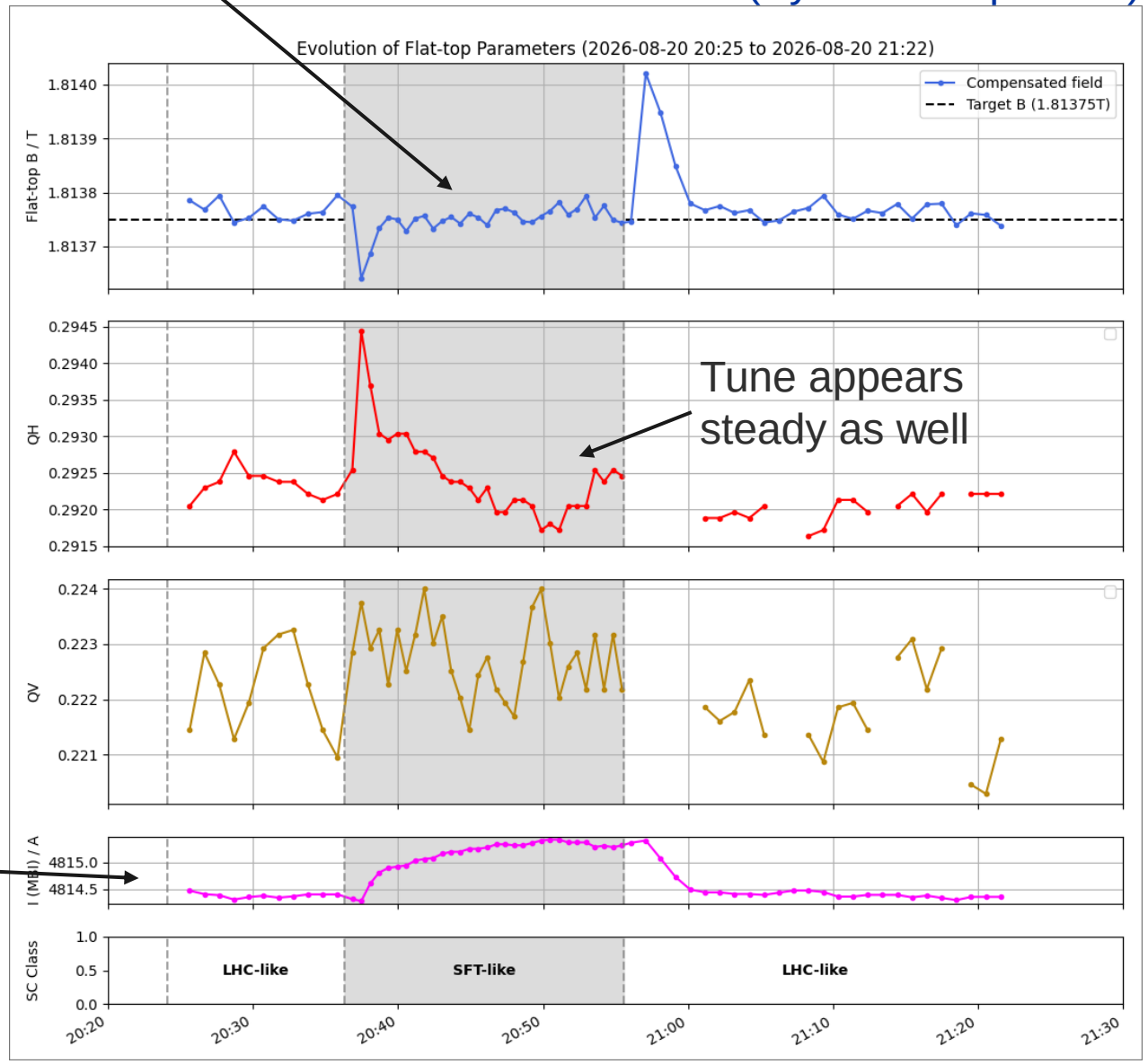
Beam Measurements

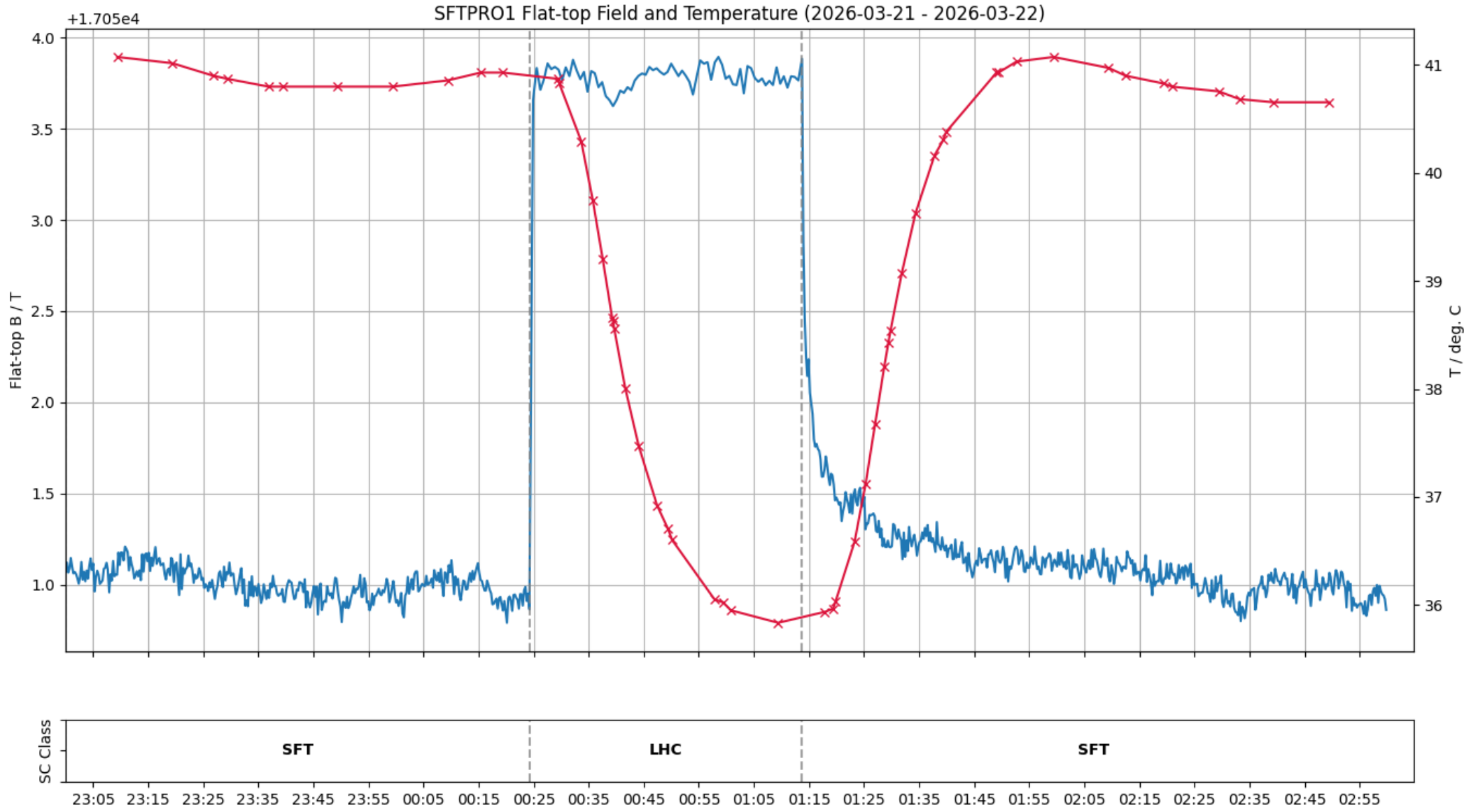
- **Compensating dipoles should let us examine pure quadrupole effects**
- **...but tune appears to be coupled to dipole hysteresis**
  - **Unexpected: not (purely) a quadrupole effect**
  - **Radial loop suspected**
- **Ran with synchro-loop at FT and compensator on: <u>appeared to restore both field and tune</u>**
- **During ions: couldn't extract**

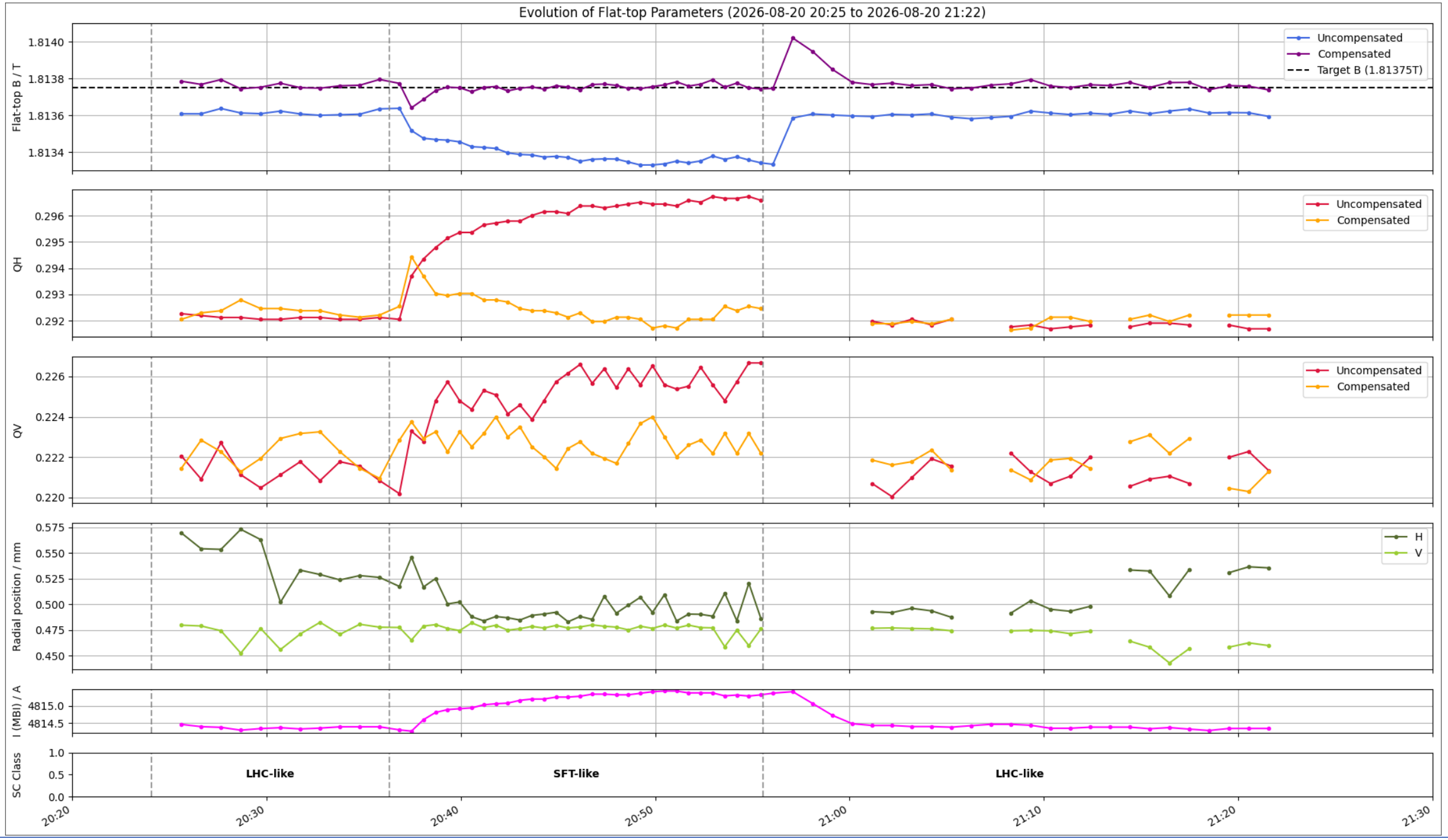
IMAINS shows action of compensator

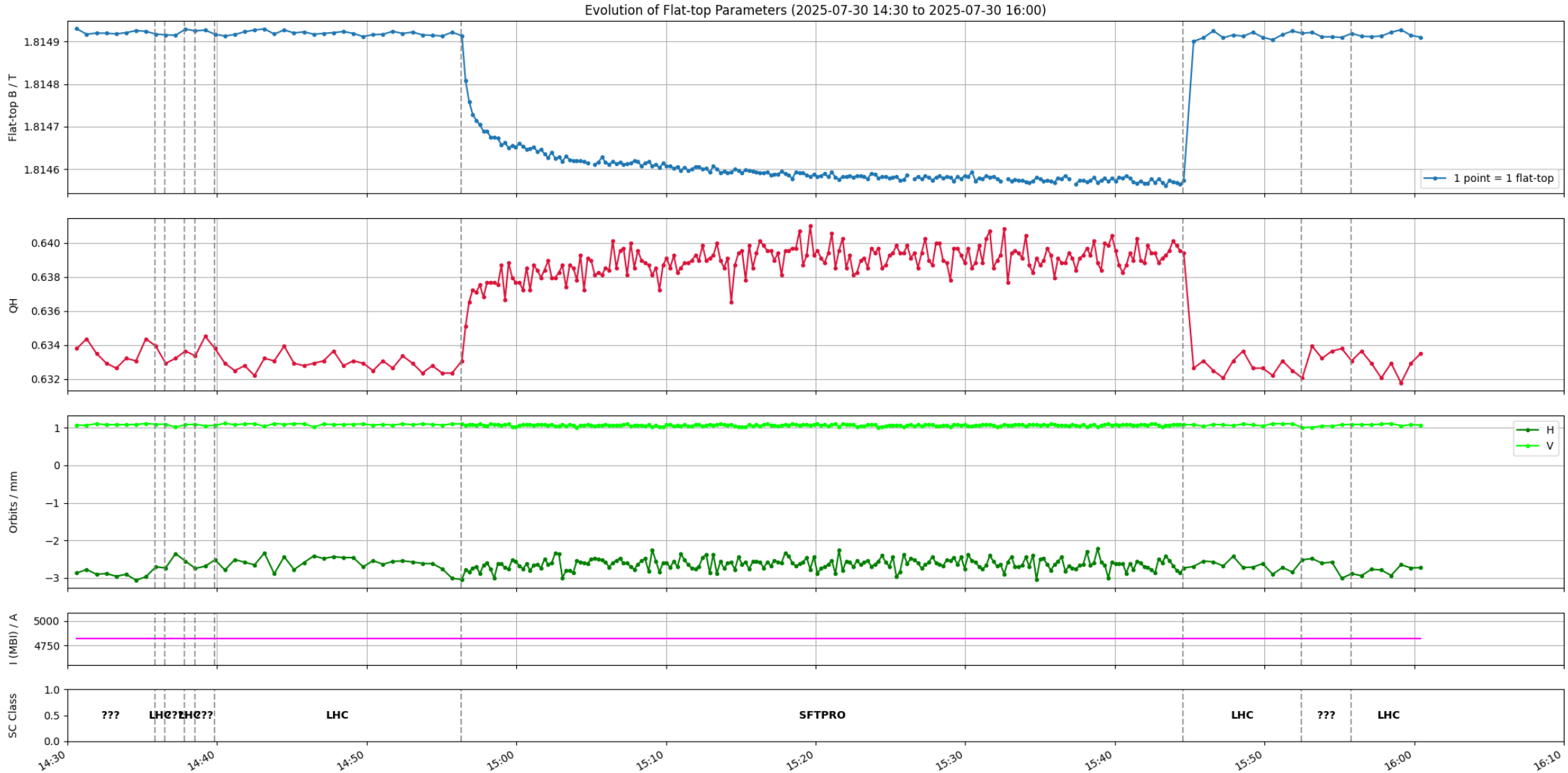
Compensator holds field steady

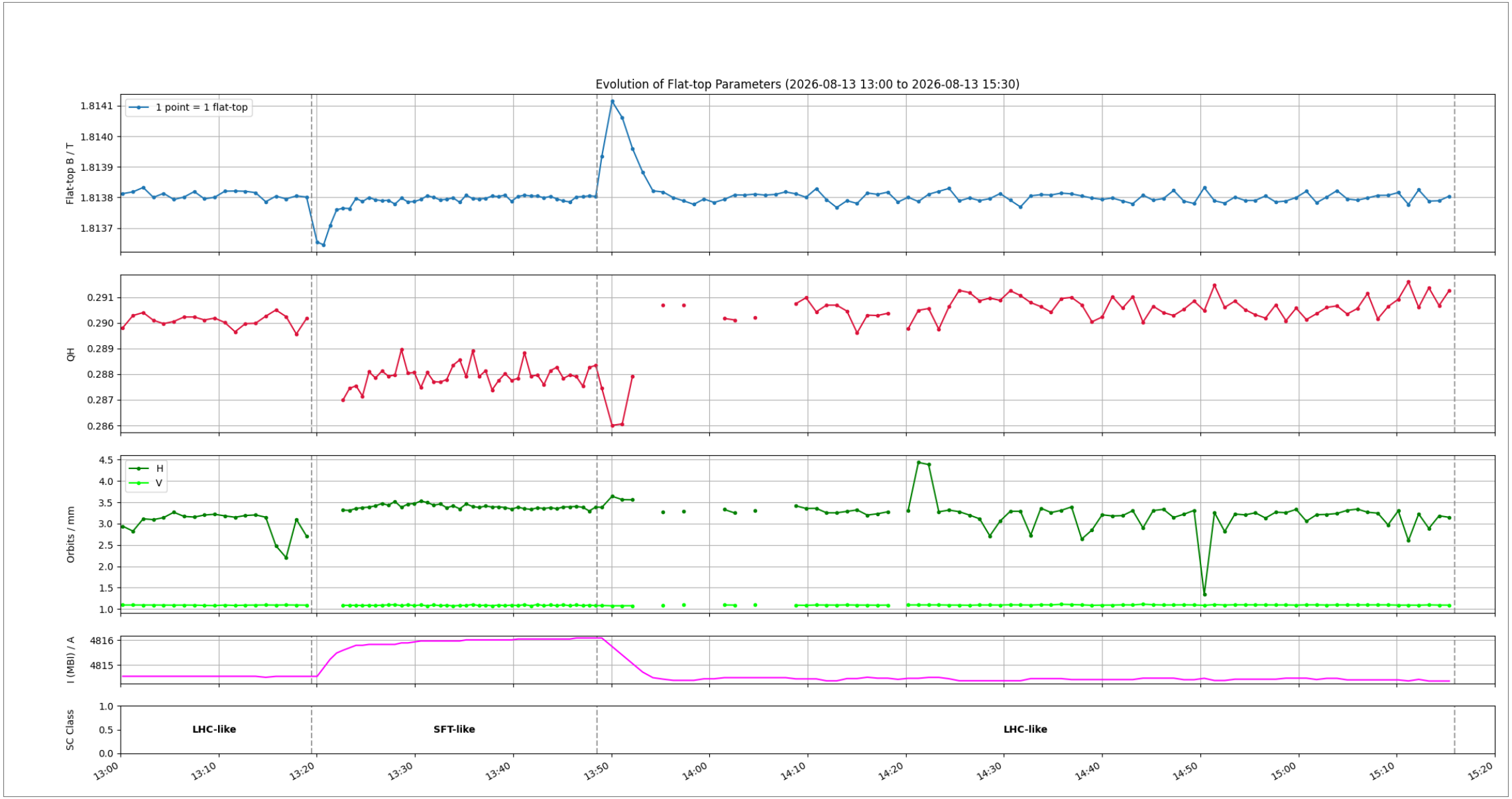
(Synchro-loop at FT)




SFTPRO1 Flat-top Field and Temperature (2026-03-21 - 2026-03-22)
+1.705e4
Flat-top B / T
T / deg. C
SC Class
SFT
LHC
SFT
23:05 23:15 23:25 23:35 23:45 23:55 00:05 00:15 00:25 00:35 00:45 00:55 01:05 01:15 01:25 01:35 01:45 01:55 02:05 02:15 02:25 02:35 02:45 02:55


Evolution of Flat-top Parameters (2026-08-20 20:25 to 2026-08-20 21:22)
Flat-top B / T
1.8140
1.8138
1.8136
1.8134
Uncompensated
Compensated
Target B (1.81375T)
QH
0.296
0.295
0.294
0.293
0.292
Uncompensated
Compensated
QV
0.226
0.224
0.222
0.220
Uncompensated
Compensated
Radial position / mm
0.575
0.550
0.525
0.500
0.475
0.450
H
V
I (MBI) / A
4815.0
4814.5
SC Class
1.0
0.5
0.0
LHC-like
SFT-like
LHC-like
20:20
20:30
20:40
20:50
21:00
21:10
21:20
21:30


Evolution of Flat-top Parameters (2025-07-30 14:30 to 2025-07-30 16:00)
Flat-top B / T
1.8149
1.8148
1.8147
1.8146
1 point = 1 flat-top
QH
0.640
0.638
0.636
0.634
0.632
Orbits / mm
1
0
−1
−2
−3
H
V
I (MBI) / A
5000
4750
SC Class
1.0
0.5
0.0
???
LHC
???
LHC
???
LHC
SFTPRO
LHC
???
LHC
14:30
14:40
14:50
15:00
15:10
15:20
15:30
15:40
15:50
16:00
16:10


Evolution of Flat-top Parameters (2026-08-13 13:00 to 2026-08-13 15:30)
1 point = 1 flat-top
Flat-top B / T
1.8141
1.8140
1.8139
1.8138
1.8137
QH
0.291
0.290
0.289
0.288
0.287
0.286
Orbits / mm
H
V
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
I (MBI) / A
4816
4815
SC Class
1.0
0.5
0.0
LHC-like
SFT-like
LHC-like
13:00
13:10
13:20
13:30
13:40
13:50
14:00
14:10
14:20
14:30
14:40
14:50
15:00
15:10
15:20